# Wireless LANs

# Wireless LANs

- Wireless Communication is one of the fastest growing technologies.
- Wireless LANs can be found on College campuses, in office buildings, and in many public areas.
- Promising Technologies for LANs-
  - IEEE 802.11 wireless LANs- called Wireless Ethernet
  - Bluetooth, Zigbee, LoRA WAN etc…

# Characteristics of wireless LANs

**Advantages**

- very flexible within the reception area
- Ad-hoc networks without previous planning possible
- (almost) no wiring difficulties (e.g. historic buildings, firewalls)
- More robust against disasters like, e.g., earthquakes, fire - or users pulling a plug...

# Contd...

Disadvantages

- Typically very low bandwidth compared to wired networks (1-10 Mbit/s)
- Many proprietary solutions, especially for higher bit-rates, standards take their time (e.g. IEEE 802.11)
- Products have to follow many national restrictions if working wireless, it takes a very long time to establish global solutions like, e.g., IMT-2000

# Design goals for wireless LANs

- Global, seamless operation
- Low power for battery use
- No special permissions or licenses needed to use the LAN
- Robust transmission technology
- Easy to use for everyone, simple management
- Protection of investment in wired networks
- Security (no one should be able to read my data), privacy (no one should be able to collect user profiles), safety (low radiation)

# IEEE 802.11

- IEEE has defined the specifications for a wireless LAN, called IEEE 802.11, which covers the physical and data link layers.
  - Architecture
  - MAC Sublayer
  - Physical Layer

***Architecture***

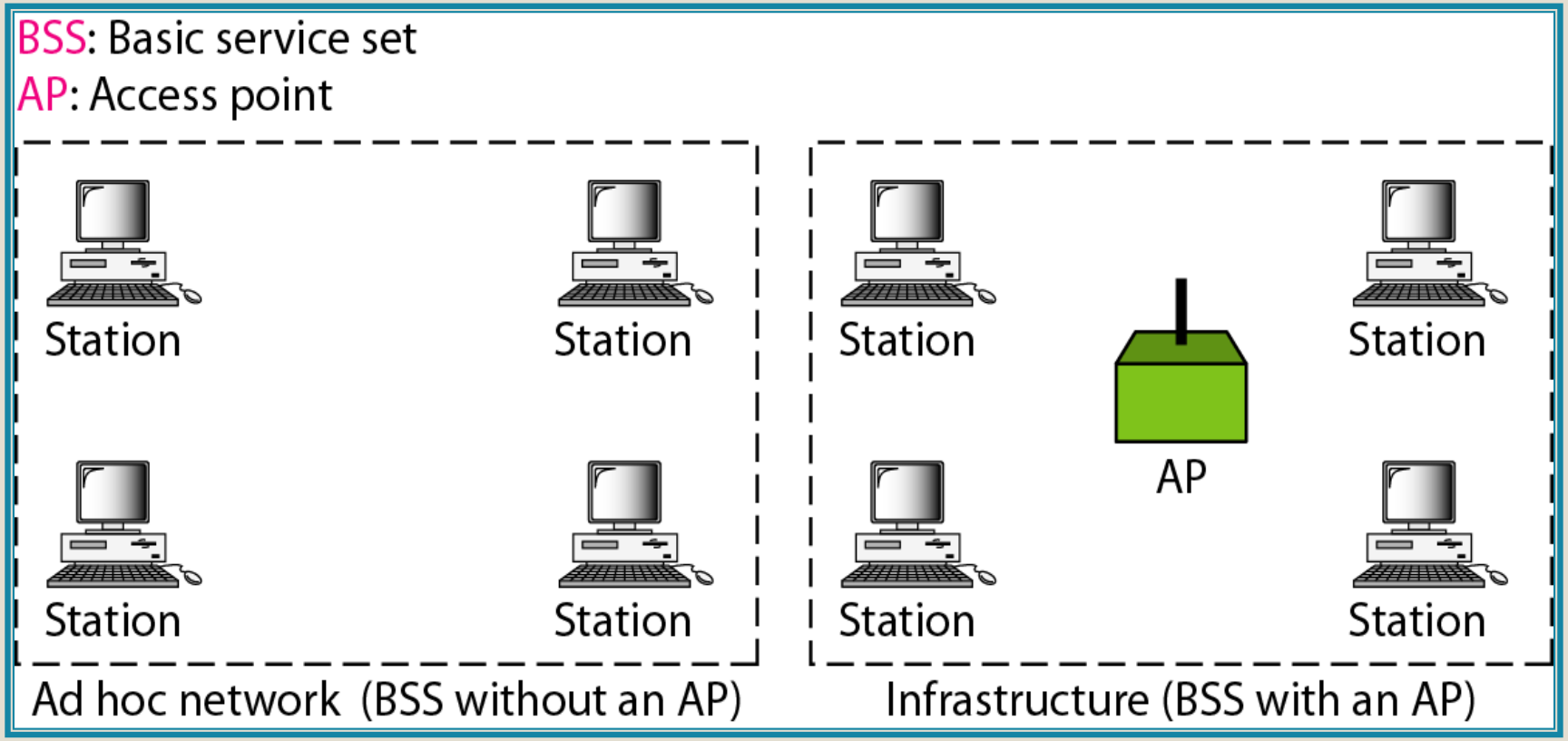
✓ Basic Service Set

✓ Extended service Set

# Basic Service Set

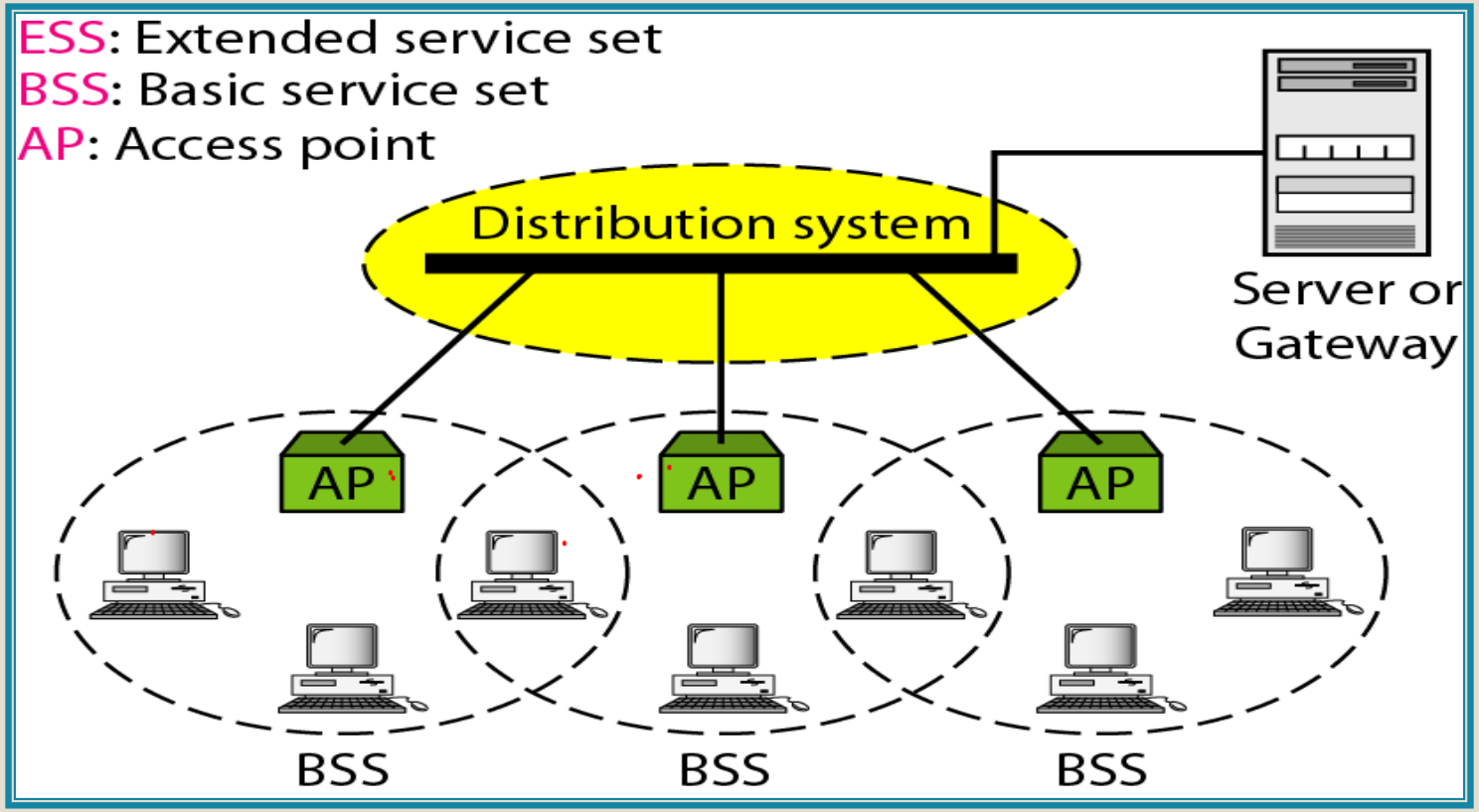
- Basic building block of wireless LAN
- Made of stationary (or) mobile wireless station and Optional central base station is known as Access Point(AP).
- A BSS without an AP is called an ad hoc network.
  - Stand alone network
  - It cannot send data to other BSSs
  - A BSS with an AP is called an infrastructure network.

# Contd...


BSS: Basic service set
AP: Access point
Station
Station
Station
Station
Station
Station
AP
Station
Station
Ad hoc network (BSS without an AP)
Infrastructure (BSS with an AP)

# Extended service Set

- Two types of stations-mobile and stationary.



# Contd…

- An extended service set (ESS) is made up of two or more BSSs with APs.
- In this case, the BSSs are connected through a *distribution system*, which is usually a wired LAN.
- The distribution system connects the APs in the BSSs.
- Two types of stations: mobile and stationary.
- The mobile stations are normal stations inside a BSS.

# Contd…

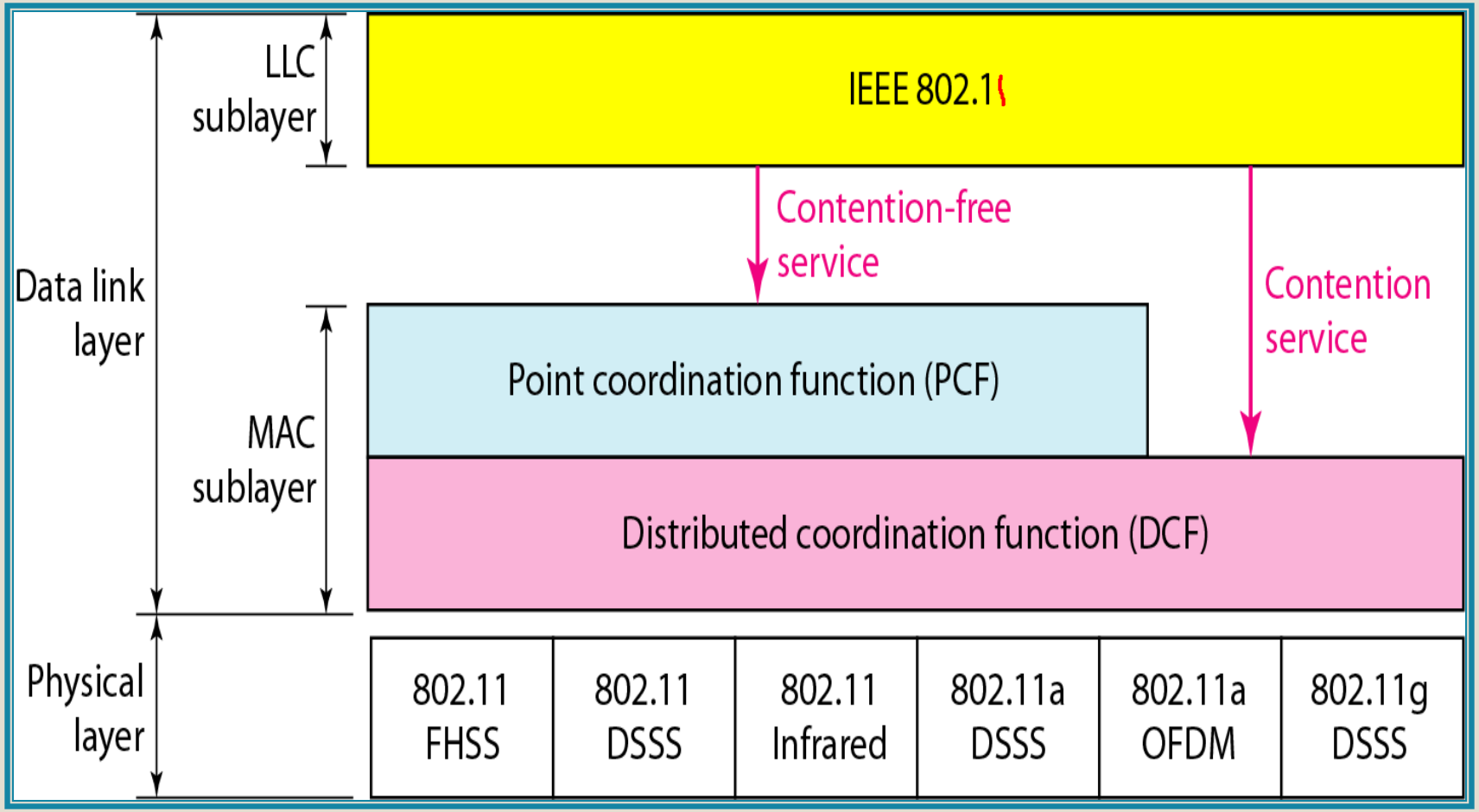
- The stationary stations are AP stations that are part of a wired LAN.
- When BSSs are connected, the stations within reach of one another can communicate without the use of an AP.
- Communication between two stations in two different BSSs usually occurs via two APs.

# Station Types

There are 3 different types of stations based on their mobility in a wireless LAN

- No Transition – It is either stationary(not moving) or moving only inside a BSS
- BSS Transition – It can move from one BSS to another, but the movement is confined inside one ESS.
- ESS Transition mobility - It can move from one ESS to another.

# MAC LAYERS IN IEEE 802.11 STANDARD



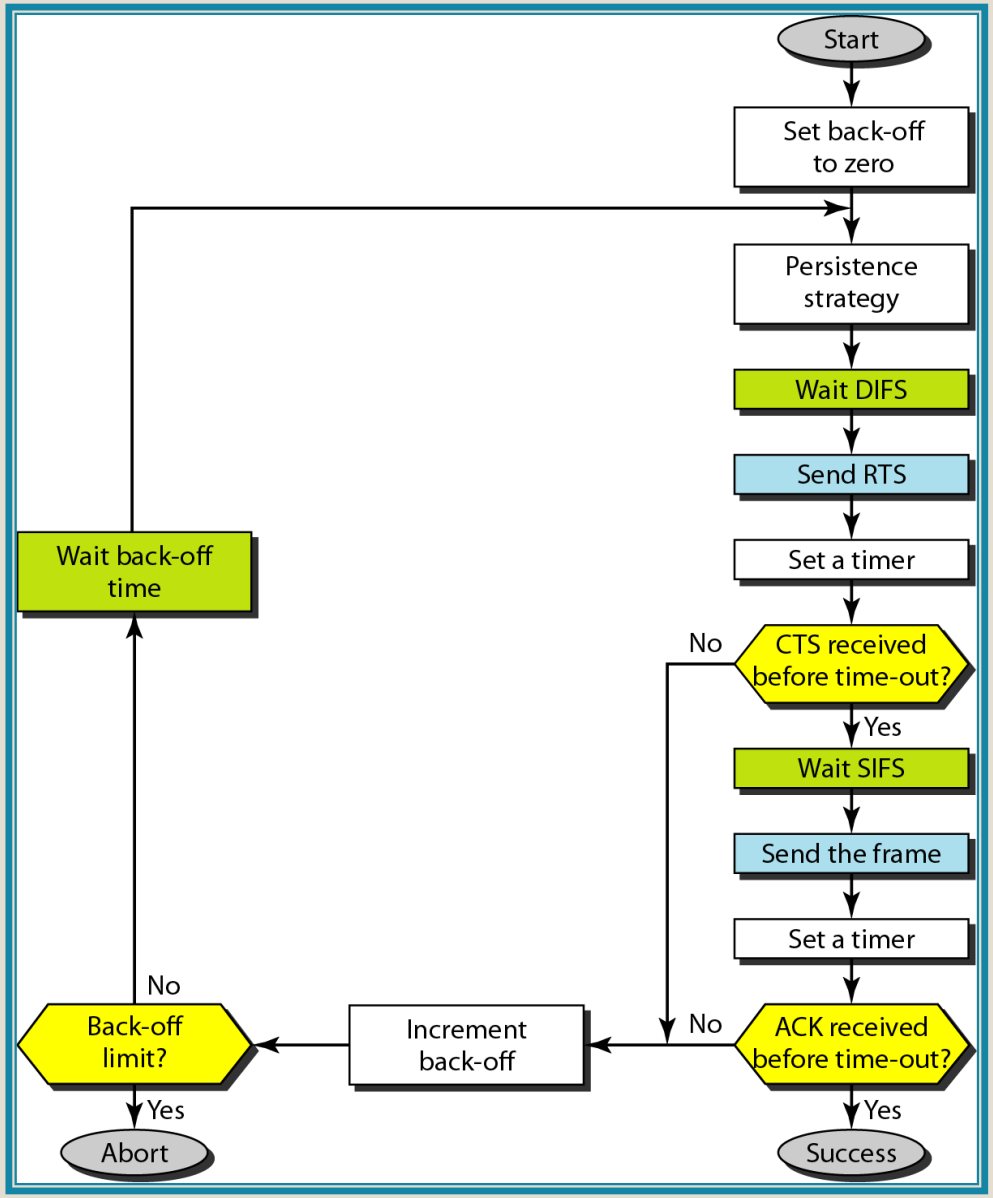
# MAC Sublayer

- IEEE 802.11 defines two MAC sublayers:
  - Distributed coordination function (DCF)
  - Point coordination function (PCF)

# Distributed Coordination Function(DCF)

- Uses CSMA/CA as the access method.
- Wireless LAN cannot implement CSMA/CD for three reasons.
  - For collision detection a station must be able to send data & receive collision signals at the same time. this can mean costly stations and increased BW requirement.
  - Collision may not be detected because of the hidden station problem.
  - The distance between stations can be great. signal fading could prevent a station at one end from hearing a collision at the other end

# CSMA/CA FLOWCHART



# Contd…

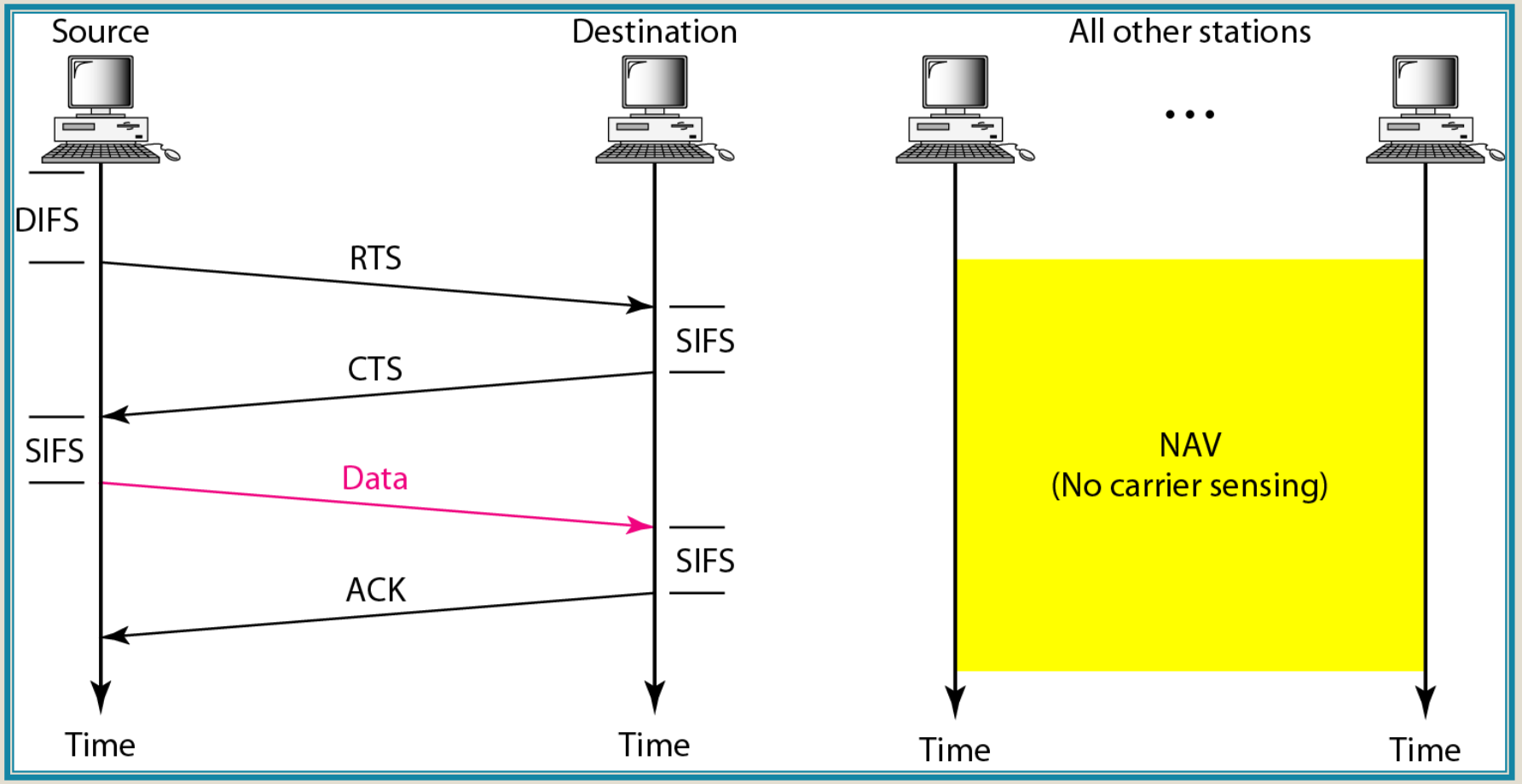
1. Before sending a frame, the source station senses the medium by checking the energy level at the carrier frequency.
   a. The channel uses a persistence strategy with back-off until the channel is idle.
   b. After the station is found to be idle, the station waits for a period of time called the **distributed interframe space (DIFS);** then the station sends a control frame called the request to send (RTS).
2. After receiving the RTS and waiting a period of time called the **short interframe space (SIFS),** the destination station sends a control frame, called the clear to send (CTS), to the source station.

# Contd...

This control frame indicates that the destination station is ready to receive data.

3. The source station sends data after waiting an amount of time equal to SIFS.
4. The destination station, after waiting an amount of time equal to SIFS, sends an acknowledgment to show that the frame has been received. Acknowledgment is needed in this protocol because the station does not have any means to check for the successful arrival of its data at the destination.

# CSMA/CA and NAV



# Network Allocation Vector(NAV)

- When a station sends an RTS frame, it includes the duration of the time that it needs to occupy the channel.
- The stations that are affected by this transmission create a timer called a Network Allocation Vector(NAV)
- It shows how much time must pass before these stations are allowed to check the channel for idleness.

# Collision during Handshaking

- Two or more stations may try to send RTS frames at the same time. These control frames may collide.
- There is no mechanism for collision detection.

# Point Coordination Function(PCF)

- Optional access method- implemented in infrastructure network (not in ad-hoc network).
- It is implement in top of DCF- used mostly in time sensitive transmission.
- It is centralized contention-free polling access method.
- The AP performs polling for stations that are capable of being polled.
- The stations are polled one after another sending any data they have to the AP.

# Fragmentation

- Wireless environment is very noisy.
- A corrupt has to be retransmitted. therefore fragmentation is recommended.

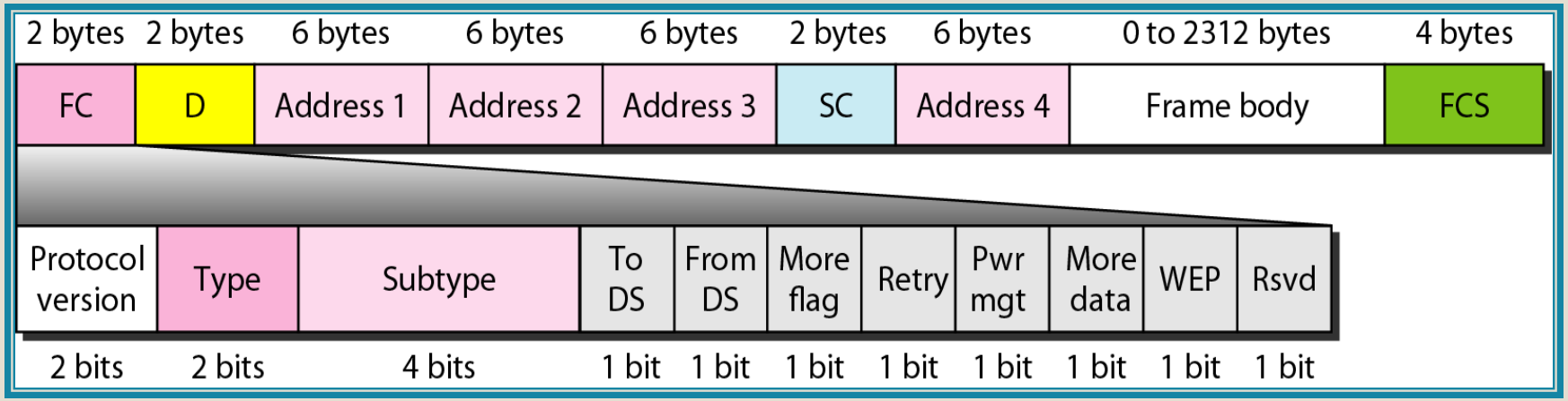
*Fragmentation*

- The division of large frames into smaller ones.
- It is more efficient to resend a small frame than a large frame.

*Frame Formats*

- Frame control- defines the type of frame and some control information
- D – defines the duration of the transmission that is used to set the value of NAV

# Frame format



# Frame format

- FC Frame control - Defines the type of frame
- D - duration of the transmission
- Address - it depends on the value of To DS and From DS subfields.
- Sequence control - defines the sequence number of frames to be used in flow control.
- Frame body – it can be between 0- 2312 bytes, contain information based on type and subtype defined in FC field.
- FCS – 4 bytes long and contains a CRC-32 error detection sequence.

# Subfields in FC field

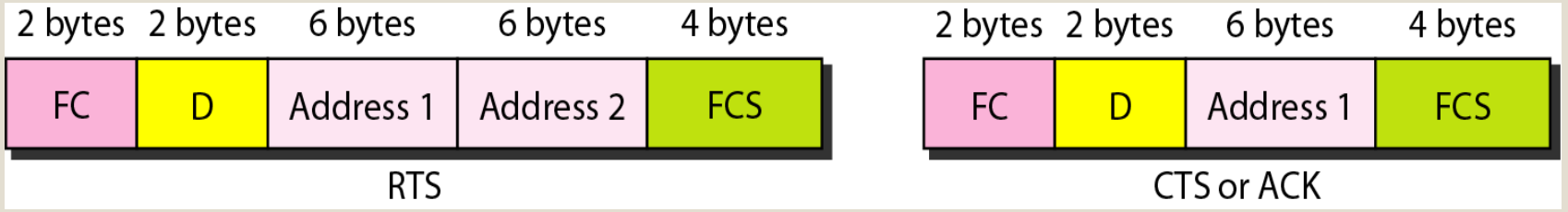
| *Field* | *Explanation* |
|---|---|
| Version | Current version is 0 |
| Type | Type of information: management (00), control (01), or data (10) |
| Subtype | Subtype of each type (see Table 14.2) |
| To DS | Defined later |
| From DS | Defined later |
| More flag | When set to 1, means more fragments |
| Retry | When set to 1, means retransmitted frame |
| Pwr mgt | When set to 1, means station is in power management mode |
| More data | When set to 1, means station has more data to send |
| WEP | Wired equivalent privacy (encryption implemented) |
| Rsvd | Reserved |

# Frame Types

Three types

- Management frames(00) – used for initial communication between stations and access points.
- Control frames(01) – used for accessing the channel and acknowledging frames.
- Data frames(10) – used for carrying data and control information.

# Frame Types



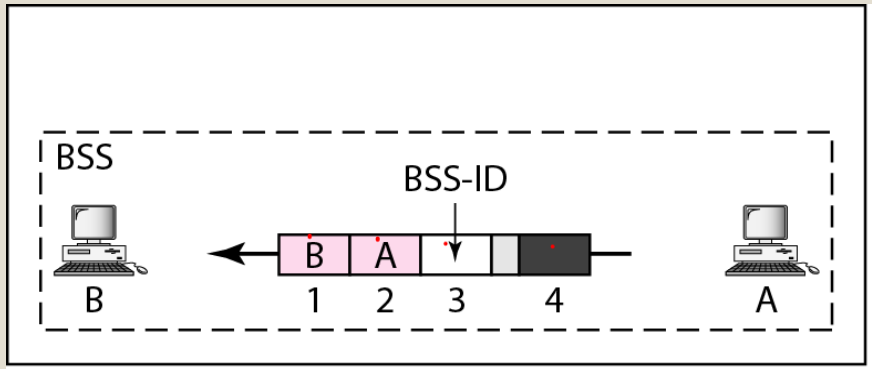

**Values of subfields in control frames (type field-01)**

| *Subtype* | *Meaning* |
|---|---|
| 1011 | Request to send (RTS) |
| 1100 | Clear to send (CTS) |
| 1101 | Acknowledgment (ACK) |

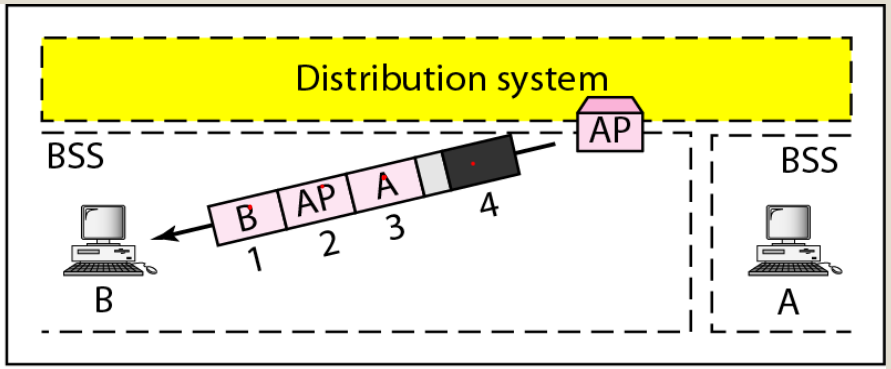
# Addresses & Addressing mechanisms

| *To DS* | *From DS* | *Address 1* | *Address 2* | *Address 3* | *Address 4* |
|---|---|---|---|---|---|
| 0 | 0 | Destination | Source | BSS ID | N/A |
| 0 | 1 | Destination | Sending AP | Source | N/A |
| 1 | 0 | Receiving AP | Source | Destination | N/A |
| 1 | 1 | Receiving AP | Sending AP | Destination | Source |

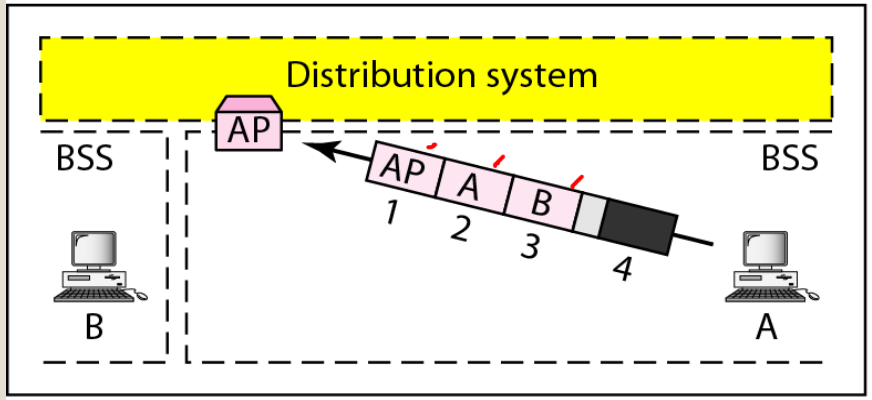
# Addresses & Addressing mechanisms



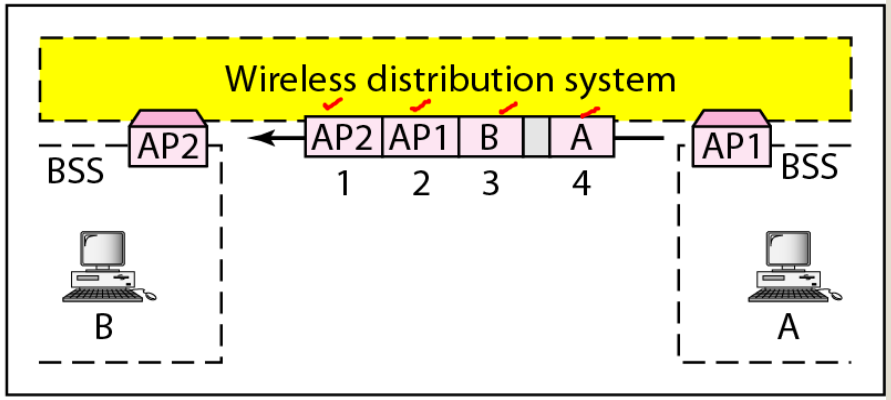

a. Case 1




b. Case 2



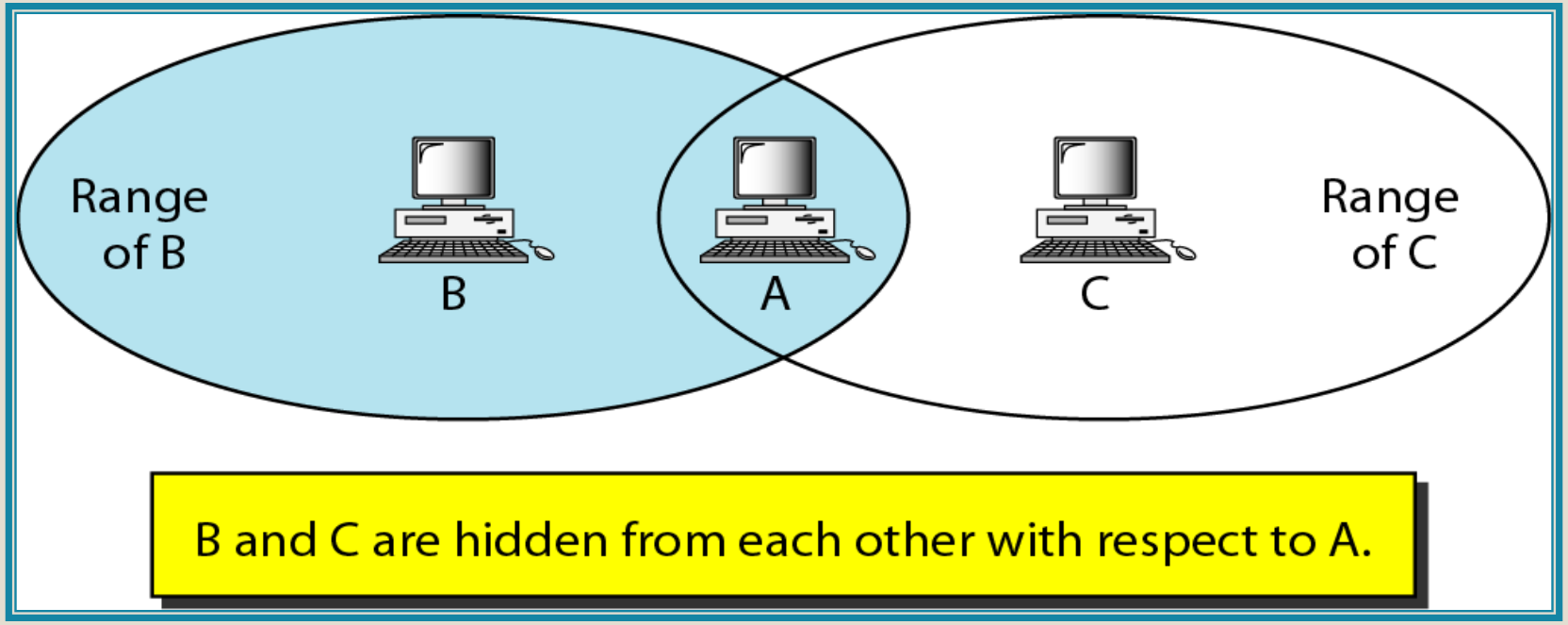

c. Case 3



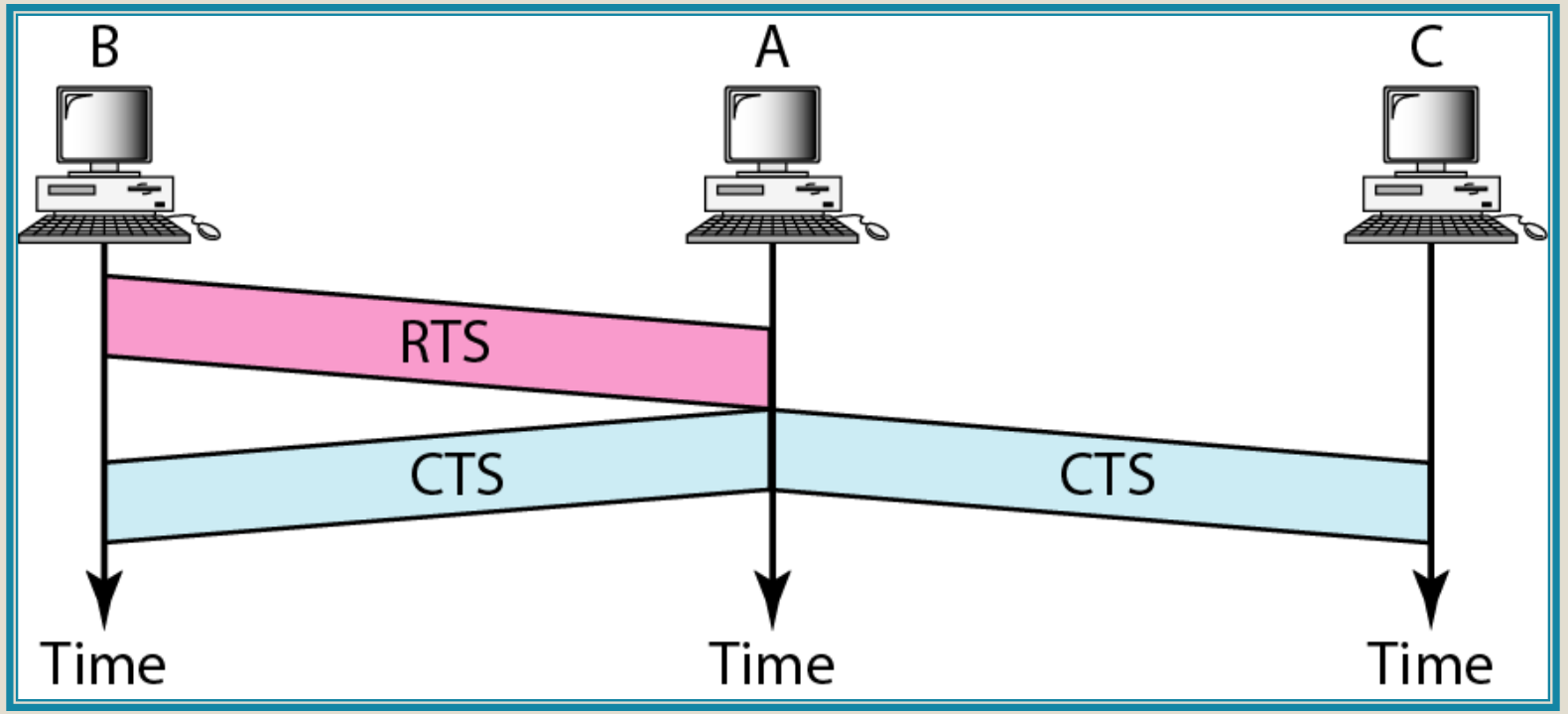

d. Case 4

- The IEEE 802.11 addressing mechanism specifies four cases, defined by the value of the two flags in the FC field, *To DS* and *From DS*.
- Each flag can be either 0 or 1, resulting in four different situations.
- Note that address 1 is always the address of the next device.
- Address 2 is always the address of the previous device.
- Address 3 is the address of the final destination station if it is not defined by address 1.
- Address 4 is the address of the original source station if it is not the same as address 2.

# Hidden station problem



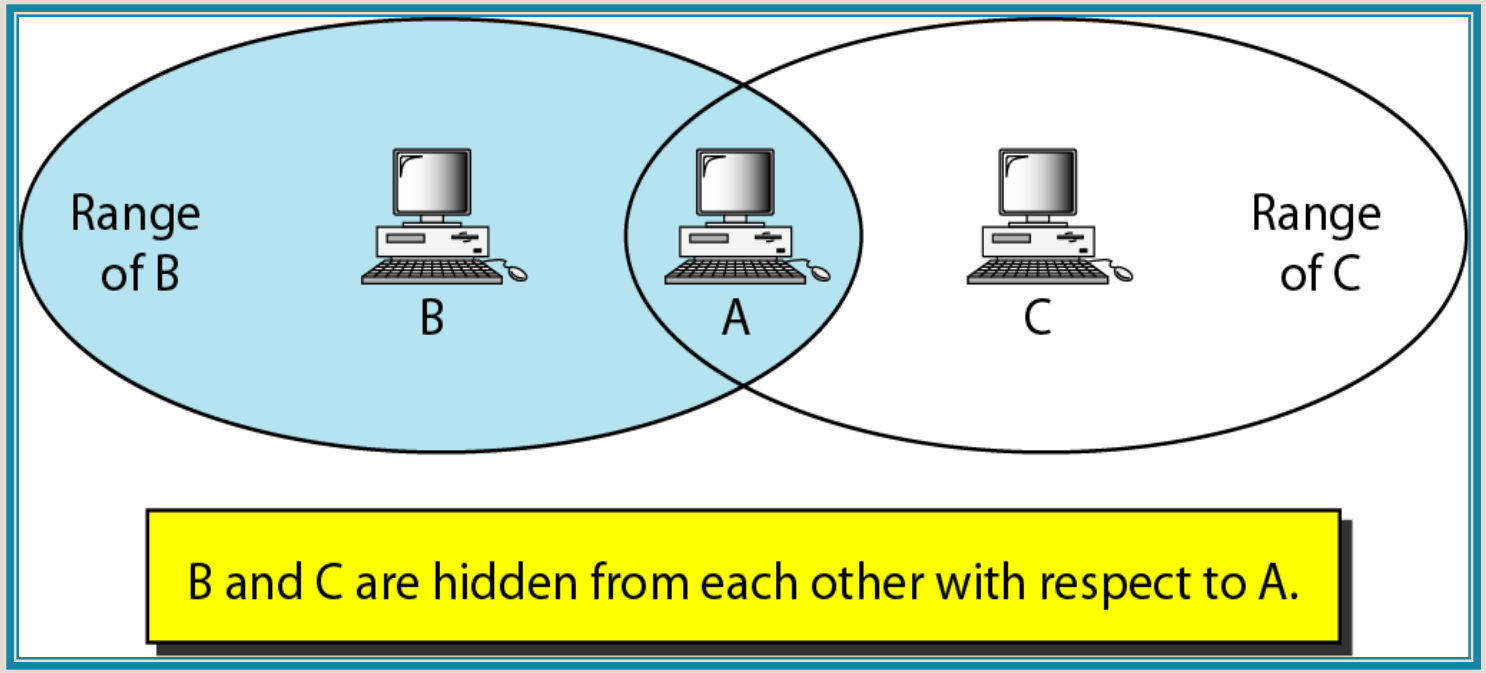
# Use of handshaking
## (to prevent hidden station problem)



# Contd...

- In wireless networks, when two terminals are not within the radio range to each other then they cannot transmit directly. This is called hidden terminal problem.
- Station B has a transmission range shown by the left oval (sphere in space); every station in this range can hear any signal transmitted by station B.
- Station C has a transmission range shown by the right oval (sphere in space); every station located in this range can hear any signal transmitted by C.
- Station C is outside the transmission range of B; likewise, station B is outside the transmission range of C.

# Contd...


Range
of B
B
A
C
Range
of C
B and C are hidden from each other with respect to A.

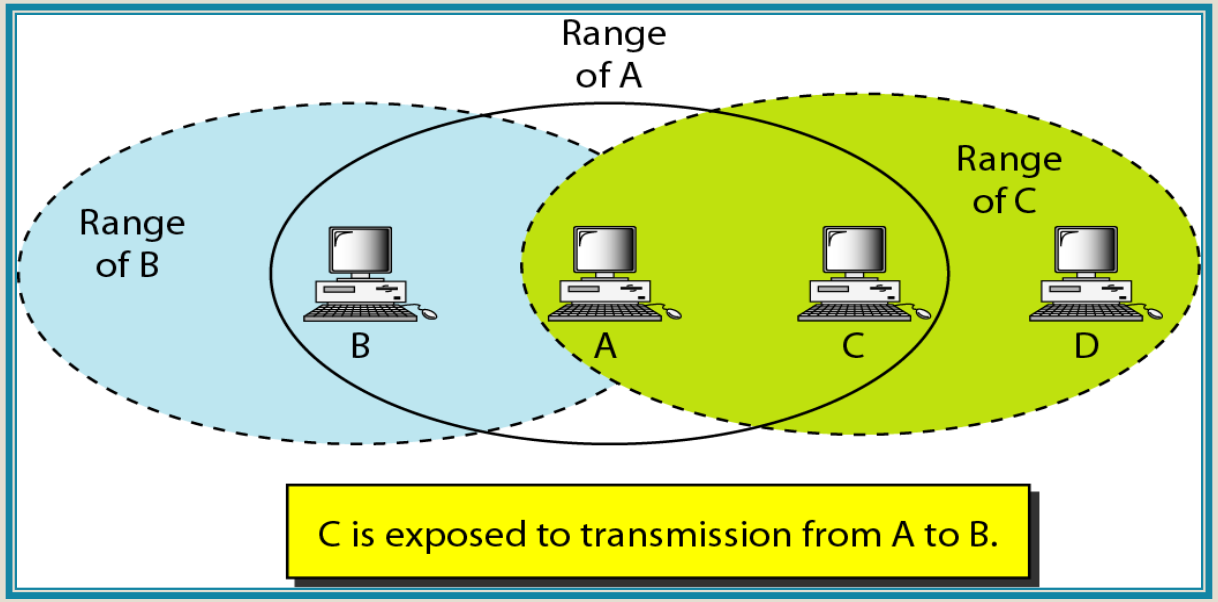
# Contd…

- Station A, however, is in the area covered by both B and C; it can hear any signal transmitted by B or C.
- Assume that station B is sending data to station A.
- In the middle of this transmission, station C also has data to send to station A.
- However, station C is out of B's range and transmissions from B cannot reach C. Therefore C thinks the medium is free.
- Station C sends its data to A, which results in a collision at A because this station is receiving data from both B and C.

# Contd...

- In this case, stations B and C are hidden from each other with respect to A.
- Hidden stations can reduce the capacity of the network because of the possibility of collision.
- The solution to the hidden station problem is the use of the handshake frames (RTS and CTS)
- RTS message from B reaches A, but not C. However, because both B and C are within the range of A, the CTS message, which contains the duration of data transmission from B to A reaches C.
- Station C knows that some hidden station is using the channel and refrains from transmitting until that duration is over.

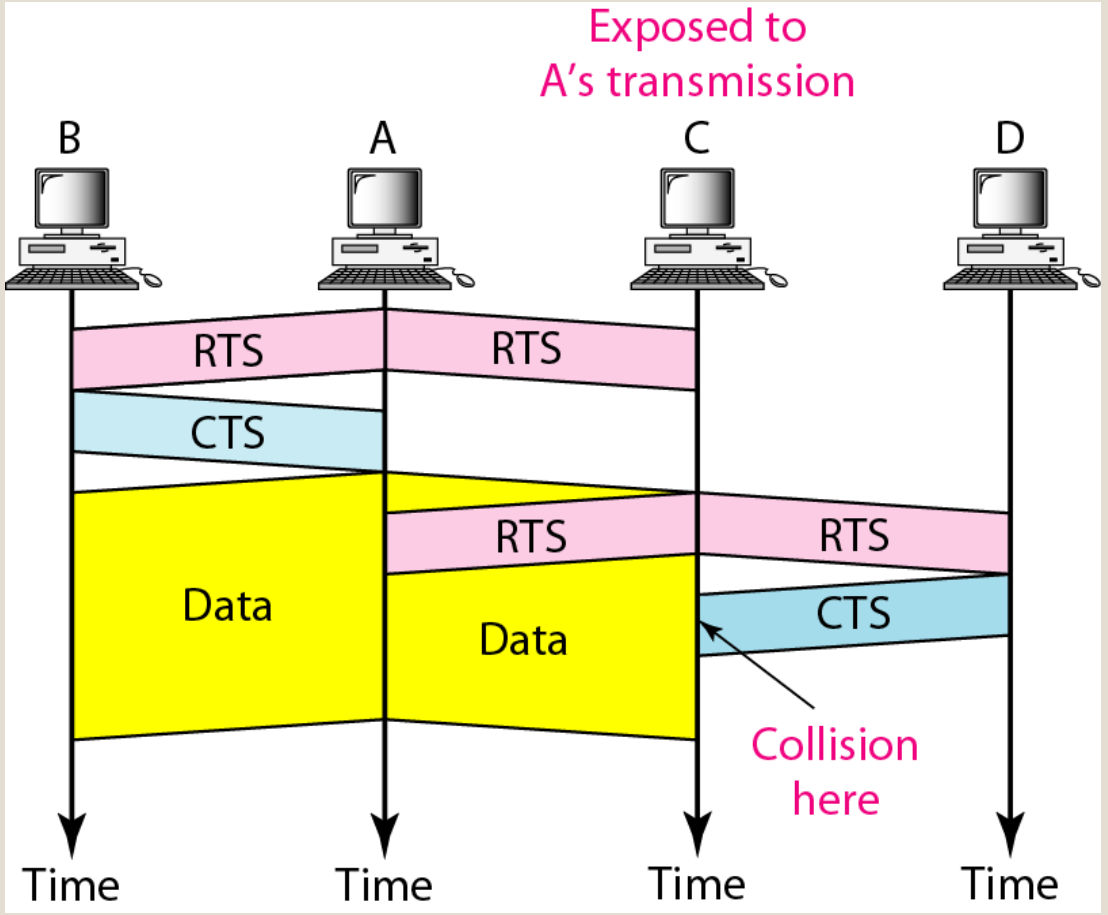
# Exposed station problem




C wants to send to D, but hears A talking to B, so assumes the medium is (incorrectly) busy.

## Use of handshaking in exposed station problem

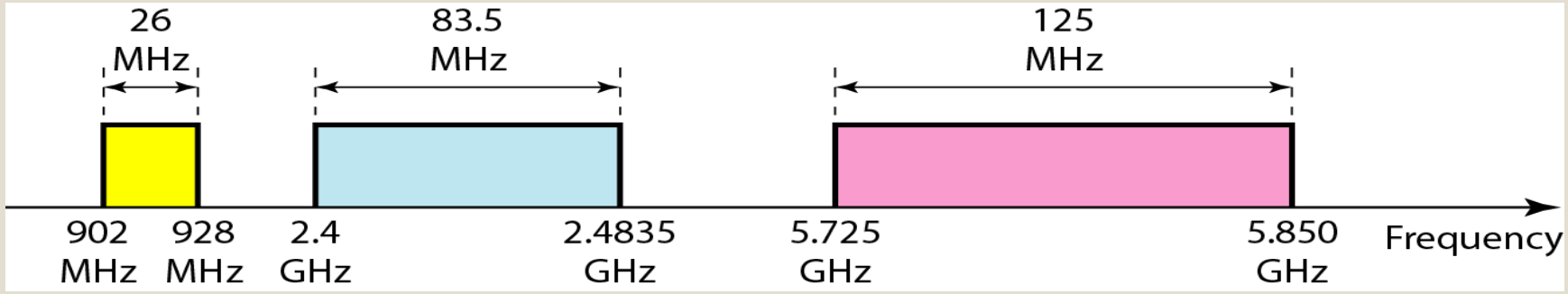
## *Use of handshaking in exposed station problem*



# Physical layers

| *IEEE* | *Technique* | *Band* | *Modulation* | *Rate (Mbps)* |
|---|---|---|---|---|
| 802.11 | FHSS | 2.4 GHz | FSK | 1 and 2 |
| | DSSS | 2.4 GHz | PSK | 1 and 2 |
| | | Infrared | PPM | 1 and 2 |
| 802.11a | OFDM | 5.725 GHz | PSK or QAM | 6 to 54 |
| 802.11b | DSSS | 2.4 GHz | PSK | 5.5 and 11 |
| 802.11g | OFDM | 2.4 GHz | Different | 22 and 54 |

## Industrial, scientific, and medical (ISM) band

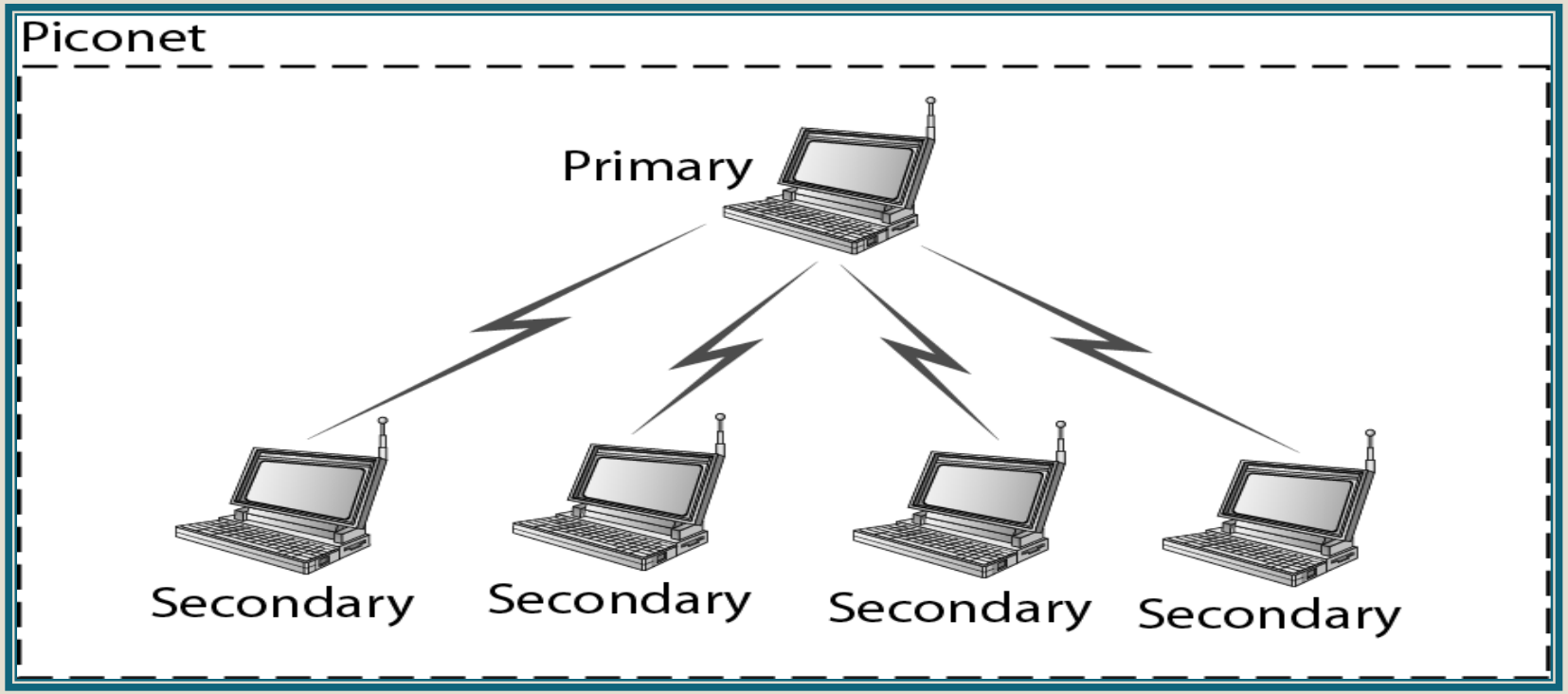


# Bluetooth

- Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers, cameras, printers, and so on.
- A Bluetooth LAN is an ad hoc network, which means that the network is formed spontaneously.
- Bluetooth was originally started as a project by Ericsson Company.

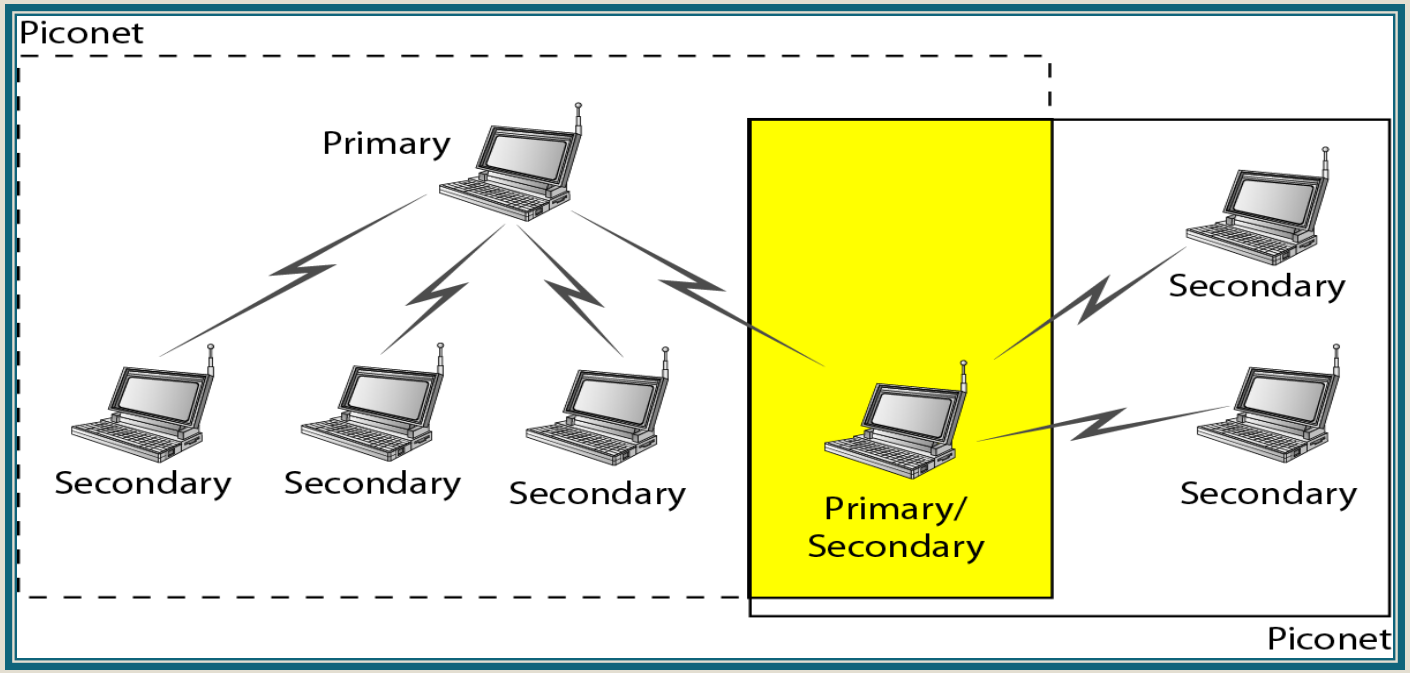
# Piconet

- Piconet - a small network.
- Eight stations- one of which is primary and rest are called secondaries
- All secondary stations synchronize their clocks and hopping sequence with the primary.



# Scatternet

- Combined form of piconet - scatternet.
- A secondary station in one piconet can be the primary in another piconet.



*Thank you*